20250527EB_MN (ATP, CellTiter Glo)

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2025年5月27日 星期二

Differentiation: BP w/ 1.5mM pyruvate.

Experiment: full T8 or Neurobasal-A

split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs

2025年5月29日 星期四

T2 -> T3

2025年5月31日 星期六

T3 -> T4

2025年6月2日 星期一

T4 -> T5

2025年6月4日 星期三

T5 -> T6

2025年6月5日 星期四

T6 -> T6 (~14mL for over-weekend)

2025年6月9日 星期一

T6 -> T7
Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) o/n on 6/6
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine over weekend on 6/7
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)

7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

coating date

	A	B
1	0.01% PLO with Na borate	2025/6/6
2	laminin	2025/6/7

cell info

	A	B	C
1	cell	passage	well
2	WT11	P16+13	x6

Table1

	A	B
1	run1	volume (mL)
2	trypsin	3
3	HI-FBS/CTWM	6
4	run2	*take some undissolved EB
5	trypsin	2
6	HI-FBS/CTWM	4

Table3

	A	B	C	D	E	F	G	H	I
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	1.13	99	5	5.65	A	24	240	12	2549
3					B	96	57	48	2421

Plate A

	1	2	3	4	5	6
A						
B						
C						
D						

Plate B

	1	2	3	4	5	6	7	8	9	10	11	12
A												
B												
C												
D												
E												
F												
G												
H												

2025年6月11日 星期三

Change T7 -> T7 for Plate A

Add extra 100uL T7 media for Plate B

2025年6月12日 星期四

T7 -> BP T8 with 1.5mM pyruvate (1mM extra pyruvate)

2025年6月15日 星期日

1/2 BP T8 -> 1/2 BP T8

2025年6月17日 星期二

1/2 BP T8 -> 1/2 BP T8

2025年6月19日 星期四

1/2 BP T8 -> 1/2 BP T8

simplified media table

	A	B	C	D	E	F	G	H
1	condition	media	pyruvate	N2(100x), B27 (50x)	GlutaMax (100x)	T8 supp	glucose	drug
2	1	NB-A	1mM				2.5	
3	2	NB-A	1mM				0	
4	3	NB-A	1mM			v	2.5	
5	4	NB-A	1mM			v	0	
6	5	NB-A	1mM	v		v	2.5	
7	6	NB-A	1mM	v		v	0	
8	7	NB-A	1mM	v	v	v	2.5	
9	8	NB-A	1mM	v	v	v	0	
10	9	NB-A	1mM	v			2.5	
11	10	NB-A	1mM	v			0	
12	11	full T8	0.5mM	v		v	2.5	
13	12	full T8	0.5mM	v		v	0	
14	13	full T8	0.5mM	v	v	v	2.5	
15	14	full T8	0.5mM	v	v	v	0	
16	15	NB-A	1mM				0	FCCP+2DG
17	16	full T8	0.5mM	v		v	0	FCCP+2DG

Condition for plate A&B1

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	condition	media	pyruvate	N2(100x), B27 (50x)	N2	B27	GlutaMax (100x)	Glutamax	T8 supp	glucose	1.1M glucose	drug	volume
2	1	NB-A	1mM							2.5	4.55		2000
3	2	NB-A	1mM							0			2000
4	3	NB-A	1mM						v	2.5	4.55		2000
5	4	NB-A	1mM						v	0			2000
6	5	NB-A	1mM	v	20	40			v	2.5	4.55		2000
7	6	NB-A	1mM	v	20	40			v	0			2000
8	7	NB-A	1mM	v	20	40	v	20	v	2.5	4.55		2000
9	8	NB-A	1mM	v	20	40	v	20	v	0			2000
10	9	NB-A	1mM	v	20	40				2.5			2000
11	10	NB-A	1mM	v	20	40				0			2000
12	11	full T8	0.5mM	v	20				v	2.5	4.55		2000
13	12	full T8	0.5mM	v	20				v	0			2000
14	13	full T8	0.5mM	v	20	40	v	20	v	2.5	4.55		2000
15	14	full T8	0.5mM	v	20	40	v	20	v	0			2000
16	15	NB-A	1mM							0		FCCP+2DG	2000
17	16	full T8	0.5mM	v					v	0		FCCP+2DG	2000

CellTiter Glo: Plate B (only do 3 point ATP stanard curve)

prepare ATP calibration

	A	B	C	D	E	F	G
1	A	B	C	D	E	F	G
2	100mM	1mM	1uM	300nM	100nM	30nM	10nM
3	dilution factor from the previous one	100	1000		10		10
4					from C		fromE
5		1uL A in 99uL media	1uL B in 999uL	150uL in 350uL media	50uL C in 450uL media	150uL in 350uL media	50uL D in 450uL media

The ATP standard readout was low last time --> use a new aliquot for the standard curve.

Condition for plate A&B

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	condition	media	pyruvate	N2(100x), B27 (50x)	N2	B27	GlutaMax (100x)	Glutamax	T8 supp	glucose	1.1M glucose	drug	volume
2	1	NB-A	1mM							2.5	4.55		2000
3	2	NB-A	1mM							0			2000
4	3	NB-A	1mM						v	2.5	4.55		2000
5	4	NB-A	1mM						v	0			2000
6	5	NB-A	1mM	v	20	40			v	2.5	4.55		2000
7	6	NB-A	1mM	v	20	40			v	0			2000
8	7	NB-A	1mM	v	20	40	v	20	v	2.5	4.55		2000
9	8	NB-A	1mM	v	20	40	v	20	v	0			2000
10	9	NB-A	1mM	v	20	40				2.5			2000
11	10	NB-A	1mM	v	20	40				0			2000
12	11	full T8	0.5mM	v	20				v	2.5	4.55		2000
13	12	full T8	0.5mM	v	20				v	0			2000
14	13	full T8	0.5mM	v	20	40	v	20	v	2.5	4.55		2000
15	14	full T8	0.5mM	v	20	40	v	20	v	0			2000
16	15	NB-A	1mM							0		FCCP+2DG	2000
17	16	full T8	0.5mM	v					v	0		FCCP+2DG	2000

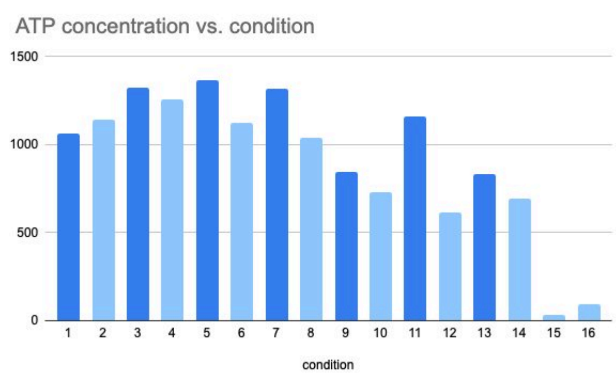
Plate B1

	1	2	3	4	5	6	7	8	9	10	11	12
A	1			9			media blank					
B	2			10			ATP C					
C	3			11			ATP E					
D	4			12			ATP G					
E	5			13								
F	6			14								
G	7			15								
H	8			16								

Table4

	A	B	C
1		9:40	prepare Celltiter glo reagent
2	-105	10:00	prepare media into a 96-well plate
3	-75	12:44	add media to cells
4	-23	13:35	prepare ATP standard and add samples
5	-17	13:50	add celltiter glo reagent
6	-12	13:55	put the plate into cytation 5 and start program to shake
7	-10	13:57	wait
8	0	14:07	measure

image.png



------------Plan for 2DG titration (save for future use)----------------

prepare media with different conc. of 2DG

	A	B	C	D	E	F	G
1	condition #	diluted from (mM)	targetconc. (mM)	dilution factor	final volume (uL)	added volume (uL)	media volume
2	A	2000	10	200	1100	5.5	1094.5
3	B	2000	6	333	1000	3.0	997.0
4	C	2000	2	1000	1000	1.0	999.0
5	D	10	0.6	17	1000	60.0	940.0
6	E	10	0.2	50	1000	20.0	980.0
7	F		0		1000	0.0	1000.0

2025年6月20日 星期五

Imaging (plate A)

Plate A_layout

	1	2	3	4	5	6
A	1	5	9			
B	2	6	10			
C	3	7	13			
D	4	8	14			

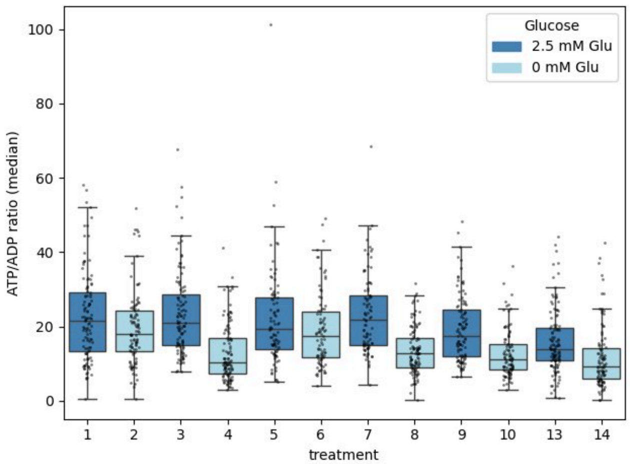
Condition for plate A&B2

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	condition	treat time	image time	media	pyruvate	N2(100x), B27 (50x)	N2	B27	GlutaMax (100x)	Glutamax	T8 supp	glucose	1.1M glucose	drug	volume
2	1	9:40	10:30	NB-A	1mM							2.5	4.55		2000
3	2	9:50	10:45	NB-A	1mM							0			2000
4	3	10:00	11:05	NB-A	1mM						v	2.5	4.55		2000
5	4	10:15	11:25	NB-A	1mM						v	0			2000
6	5	10:30	11:42	NB-A	1mM	v	20	40			v	2.5	4.55		2000
7	6	10:55	11:58	NB-A	1mM	v	20	40			v	0			2000
8	7	11:10	12:13	NB-A	1mM	v	20	40	v	20	v	2.5	4.55		2000
9	8	11:30	12:30	NB-A	1mM	v	20	40	v	20	v	0			2000
10	9	11:45	12:49	NB-A	1mM	v	20	40				2.5			2000
11	10	12:03	13:04	NB-A	1mM	v	20	40				0			2000
12	11			full T8	0.5mM	v	20				v	2.5	4.55		2000
13	12			full T8	0.5mM	v	20				v	0			2000
14	13	12:19	13:29	full T8	0.5mM	v	20	40	v	20	v	2.5	4.55		2000
15	14	12:33	13:38	full T8	0.5mM	v	20	40	v	20	v	0			2000
16	15			NB-A	1mM							0		FCCP+2DG	2000
17	16			full T8	0.5mM	v					v	0		FCCP+2DG	2000

O.C. , 60X, 600x900, 16-bit

	A	B	C	D
1	channel	bit	power (%)	exposure time
2	488	16	20	200
3	Dapi-ex-Green-em	16	20	300

image.png


Glucose
2.5 mM Glu
0 mM Glu
ATP/ADP ratio (median)
0
20
40
60
80
100
1
2
3
4
5
6
7
8
9
10
13
14
treatment